BK1210 - 2 A Synchronous Step-Down Converter

Aerix Semiconductor. Datasheet Rev 2.0.

The BK1210 is a 1.2 MHz synchronous buck converter that delivers up to 2 A of output current from a 3 V to 17 V input. The output voltage is adjustable from 0.6 V using an external resistor divider.

Features: input voltage 3 V to 17 V; output current up to 2 A continuous; switching frequency 1.2 MHz fixed; feedback reference 0.6 V +/-1.5%; internal soft start 1 ms; power-good output (PGOOD, open drain); efficiency up to 94%; package SOT-563.

Parameter	Value
VIN	3 - 17 V
IOUT	2 A
fSW	1.2 MHz
VFB	0.6 V

Absolute maximum ratings and pinout

Absolute maximum ratings: VIN to GND -0.3 V to 18 V. SW to GND -0.3 V to VIN + 0.3 V. FB, EN, PGOOD to GND -0.3 V to 6 V. Junction temperature 150 C.

Pinout (SOT-563): pin 1 FB, pin 2 EN, pin 3 VIN, pin 4 GND, pin 5 SW, pin 6 PGOOD.

Setting the output voltage

VOUT = 0.6 V x (1 + R1 / R2), where R1 is the top resistor from VOUT to FB and R2 is the bottom resistor from FB to GND.

For 1.2 V output use R1 = R2 = 100 kOhm. For 3.3 V output use R1 = 453 kOhm and R2 = 100 kOhm.

Use 1% resistors. Resistor tolerance directly affects output accuracy.

Component selection

Inductor: 2.2 uH recommended for VOUT below 1.8 V, 4.7 uH for VOUT of 3.3 V and above. Saturation current should be at least 3 A. DCR below 50 mOhm.

Output capacitor: 22 uF X5R or X7R ceramic minimum. Two 22 uF in parallel reduce ripple further. Input capacitor: 10 uF ceramic close to VIN.

Output ripple with 2.2 uH and 22 uF is typically 10 mV peak-to-peak at 1 A load.

Layout

Keep the SW node area small, it is the main source of EMI. Place the input capacitor as close as possible to VIN and GND. Route the FB trace away from the inductor and SW node.